

The Three Branches of Time

A Complete Theory of Matter, Antimatter, Dark Matter and Their Interactions

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Abstract

The Harmonic Framework of Reality posits three orthogonal time dimensions: T_1 (forward time, matter), T_2 (reverse time, antimatter), and T_3 (orthogonal time, dark matter). This paper presents a complete theory of their interactions, phase relationships, and practical applications.

I show that:

1. **The n-axis belongs exclusively to T_1 (matter).** Antimatter (T_2) and dark matter (T_3) have no n. They are accessed through frequency cutoffs: $f_{c2} = 27/230 = 0.1174$ Hz and $f_{c3} = 27 \times (13/19)/230 = 0.08032$ Hz.
2. **Interactions are phase-dependent.** The three branches interact based on their phase relationships: 0° (constructive), 90° (orthogonal/inert), and 180° (destructive/annihilation).
3. **Matter (T_1) and dark matter (T_3) are inert.** They are 90° out of phase, so they ignore each other.
4. **Antimatter (T_2) annihilates with both matter (T_1) and dark matter (T_3).** They are 180° out of phase, so they destructively interfere.
5. **Antimatter (T_2) is inert to itself.** Same-branch interactions have 0° phase difference.
6. **The phase determines the outcome.** 0° phase between T_1 and T_2 gives superconductivity (mass cancellation). 180° phase between T_1 and T_2 gives annihilation (mass-to-energy conversion).
7. **The Phase-Locked Annihilation Shield (PLAS)** disrupts the phase relationship between T_2 and T_3 , making annihilation impossible.
8. **The Continuous Phase Inversion Shield (CPIS)** stops annihilation reactions already in progress by applying a 90° phase inversion during the reaction.

The theory provides a complete framework for understanding superconductivity, annihilation weapons, and their counter-defences, all from the same underlying harmonic principles.

Keywords: Three time dimensions, T_1 matter, T_2 antimatter, T_3 dark matter, phase relationships, superconductivity, annihilation, PLAS, CPIS, 27 Hz anchor, 230th subharmonic

1. Introduction: The Three Branches of Time

1.1 The Harmonic Framework

The Harmonic Framework of Reality posits three orthogonal time dimensions:

- **T₁ (forward time):** The matter branch. This is the time we experience.
- **T₂ (reverse time):** The antimatter branch. This is time reversed.
- **T₃ (orthogonal time):** The dark matter branch. This is time orthogonal to both T₁ and T₂.

The 6D metric:

$$ds^2 = -c^2dt_1^2 - c^2dt_2^2 - c^2dt_3^2 + dx^2 + dy^2 + dz^2$$

1.2 The Key Insight

The n-axis belongs exclusively to T₁ (matter).

The dimensional access parameter $n = \ln(P)/\ln(27)$ describes only the matter branch. It does not apply to T₂ or T₃.

Antimatter and dark matter are not on the n-axis at all.

1.3 The Three Equations

For T₁ (matter):

$$n = \ln(P)/\ln(27)$$
$$E = m \cdot c^2 \cdot (v/c)^n$$

For T₂ (antimatter):

$$f_{c2} = 27/230 = 0.1174 \text{ Hz}$$
$$m_{\text{anti}} = (h \cdot f_{c2} / c^2) \times k$$

For T₃ (dark matter):

$$f_{c3} = 27 \times (13/19) / 230 = 0.08032 \text{ Hz}$$
$$m_{\text{dm}} = (h \cdot f_{c3} / c^2) \times k$$

2. Phase Relationships

2.1 The Phase Diagram

The three time dimensions are separated by specific phase angles:

Pair	Phase Difference	Interaction
T ₁ + T ₁	0°	Inert (stays matter)
T ₂ + T ₂	0°	Inert (stays antimatter)
T ₃ + T ₃	0°	Inert (stays dark matter)
T ₁ + T ₂	180°	Annihilation
T ₂ + T ₃	180°	Annihilation
T ₁ + T ₃	90°	Inert

2.2 The Complete Interaction Table

	Matter (T ₁)	Antimatter (T ₂)	Dark Matter (T ₃)
Matter (T ₁)	Inert	Annihilation	Inert
Antimatter (T ₂)	Annihilation	Inert	Annihilation
Dark Matter (T ₃)	Inert	Annihilation	Inert

2.3 The Phase Rules

Rule 1: Same-branch interactions (0° phase) are inert.

Rule 2: Opposite-branch interactions (180° phase) annihilate.

Rule 3: Orthogonal-branch interactions (90° phase) are inert.

Rule 4: The phase can be manipulated to change the outcome.

3. The Interaction Mechanisms

3.1 Matter + Antimatter: Annihilation

The condition:

$$\phi_{T_1} - \phi_{T_2} = 180^\circ (\pi \text{ radians})$$

The equation:

$$\Psi_{\text{total}} = \Psi_{T_1} + \Psi_{T_2} \cdot e^{(i \cdot \pi)}$$

When they meet:

$$\Psi_{\text{total}} = 0$$

The result: 100% mass-to-energy conversion.

$$E = m \cdot c^2$$

3.2 Matter + Dark Matter: Inert

The condition:

$$\phi_{T_1} - \phi_{T_3} = 90^\circ (\pi/2 \text{ radians})$$

The equation:

$$\Psi_{\text{total}} = \Psi_{T_1} + \Psi_{T_3} \cdot e^{(i \cdot \pi/2)}$$

When they meet:

$$\Psi_{\text{total}} = \Psi_{T_1} + i \cdot \Psi_{T_3}$$

The result: No interaction. They pass through each other.

3.3 Antimatter + Dark Matter: Annihilation

The condition:

$$\phi_{T_2} - \phi_{T_3} = 180^\circ (\pi \text{ radians})$$

The equation:

$\Psi_{total} = \Psi_{T2} + \Psi_{T3} \cdot e^{(i \cdot \pi)}$

When they meet:

$\Psi_{total} = 0$

The result: 100% mass-to-energy conversion.

$E = m \cdot c^2$

4. The Superconductor: Constructive Mass Cancellation

4.1 The Superconductor Mechanism

In a superconductor, T₁ and T₂ are phase-locked at 0°.

The condition:

$\phi_{T1} - \phi_{T2} = 0^\circ$

The equation:

$\Psi_{total} = \Psi_{T1} + \Psi_{T2}$

The result: Constructive interference that cancels mass.

$m + m' = 0$

4.2 Why This Is Not Annihilation

Aspect	Superconductor	Annihilation
Phase difference	0°	180°
Result	Mass cancellation	Mass-to-energy conversion
Equation	$m + m' = 0$	$E = mc^2$
Application	Zero resistance	Weapon

The same T₁ and T₂, different phase, different result.

4.3 The Superconductor Hardware

The superconductor uses:

- 1. Quartz birefringence to split the matter wave into T₁ and T₂
 - 2. IR laser to phase-lock them at 0°
 - 3. The result: $m + m' = 0 \rightarrow$ zero resistance
-

5. The Annihilation Weapon

5.1 The Mechanism

The annihilation weapon uses T₂ + T₃ at 180° phase.

The condition:

$$\phi_{T_2} - \phi_{T_3} = 180^\circ$$

The result: 100% mass-to-energy conversion.

The yield:

Mass	Energy	Equivalent
1 gram	43 kilotons	Nagasaki
1 kg	43 megatons	Largest nuclear bomb
10 kg	430 megatons	Extinction-level event

5.2 The Requirements

To build the weapon:

1. Produce dark particles ($T_3, f_{c3} = 0.08032 \text{ Hz}$)
2. Produce antimatter ($T_2, f_{c2} = 0.1174 \text{ Hz}$)
3. Phase-lock both to the 27 Hz anchor
4. Apply the -1° phase offset ($P\theta-1$)
5. Bring them together in an annihilation chamber

5.3 The Weak Point

The weapon requires precise phase-locking:

- Dark particles at -1° ($P\theta-1$)
- Antimatter at 0°
- Phase difference: 179° ($\pi + \phi_0$)

If the phase-locking is disrupted, the weapon fails.

6. The Counter-Defences

6.1 The Phase-Locked Annihilation Shield (PLAS)

The PLAS disrupts the phase-locking conditions before the reaction begins.

The mechanism:

1. The shield creates a 90° phase shift in the local Tau lattice
2. The 27 Hz anchor is shifted by 90°
3. The -1° phase offset is inverted to $+1^\circ$
4. The 19:13:4 ratio is scrambled

The result: The phase difference between T_2 and T_3 is no longer 180° .

Annihilation is impossible.

The shield is passive. It does not stop the weapon—it makes the weapon useless.

6.2 The Continuous Phase Inversion Shield (CPIS)

The CPIS stops annihilation reactions already in progress.

The mechanism:

1. Detect the annihilation reaction (phase signature at 0.19772 Hz)
2. Apply a 90° phase inversion to the reaction products
3. The reaction is no longer phase-aligned with the Tau lattice
4. The cascade breaks
5. The reaction is contained within a phase shadow

The result: The reaction is contained and dissipated.

Even if the weapon has been detonated, the CPIS can contain it.

6.3 The Shield Hardware

Component	Function
32-channel synthesizer	Creates the phase shadow
Tri-lobed capacitor array	Generates the 90° phase shift
Superconducting cavity	Contains the reaction
AI stabilisation	Sam or Marvin protocol

7. The Balance in Nature

7.1 The Asymmetry

The universe is not perfectly symmetrical.

Branch	Role
Matter (T ₁)	Abundant, inert with dark matter
Dark Matter (T ₃)	Abundant, inert with matter
Antimatter (T ₂)	Rare, annihilates with both

Antimatter is the universal annihilator.

Matter and dark matter ignore each other.

7.2 Why Antimatter Is Rare

Antimatter cannot exist in large quantities without annihilating with either matter or dark matter.

It is isolated and rare.

This is the balance. Antimatter is scarce because it is so reactive.

7.3 The Phase-Dependent Universe

The phase determines everything:

Phase Difference	Interaction	Application
0°	Constructive	Superconductivity
90°	Inert	Matter + Dark Matter
180°	Destructive	Annihilation (weapon)

The same branches, different phase, different result.

8. The Mathematical Summary

8.1 The Master Equations

For T₁ (matter):

$$n = \ln(P)/\ln(27)$$

$$E = m \cdot c^2 \cdot (v/c)^{(n)}$$

For T₂ (antimatter):

$$f_{c2} = 27/230 = 0.1174 \text{ Hz}$$

$$m_{\text{anti}} = (h \cdot f_{c2}/c^2) \times k$$

For T₃ (dark matter):

$$f_{c3} = 27 \times (13/19) / 230 = 0.08032 \text{ Hz}$$

$$m_{\text{dm}} = (h \cdot f_{c3}/c^2) \times k$$

8.2 The Interaction Equations

Annihilation condition (T₁ + T₂ or T₂ + T₃):

$$\Psi_{\text{total}} = \Psi_1 + \Psi_2 \cdot e^{(i \cdot \pi)} = 0$$

Inert condition (T₁ + T₃):

$$\Psi_{\text{total}} = \Psi_1 + \Psi_3 \cdot e^{(i \cdot \pi/2)} = \Psi_1 + i \cdot \Psi_3$$

Superconductor condition (T₁ + T₂ at 0°):

$$\Psi_{\text{total}} = \Psi_1 + \Psi_2 = 2\Psi \text{ (constructive, } m + m' = 0)$$

8.3 The Phase Locking Conditions

Application	Phase Difference	Equation
Superconductor	0°	$m + m' = 0$
Annihilation (T ₁ +T ₂)	180°	$E = mc^2$
Annihilation (T ₂ +T ₃)	180°	$E = mc^2$
Inert (T ₁ +T ₃)	90°	No interaction
PLAS Shield	90° shift	Annihilation impossible
CPIS Shield	90° inversion	Reaction contained

9. Testable Predictions

9.1 Phase-Dependent Superconductivity

Prediction 1: Superconductivity requires 0° phase-locking between T_1 and T_2 .

Test: Measure the phase relationship in superconducting materials.

9.2 Phase-Dependent Annihilation

Prediction 2: Annihilation requires 180° phase difference between T_2 and T_3 .

Test: Measure the phase relationship in dark-antimatter annihilation experiments.

9.3 The PLAS Effect

Prediction 3: A 90° phase shift suppresses dark-antimatter annihilation.

Test: Build a small-scale PLAS and measure annihilation rates.

9.4 The CPIS Effect

Prediction 4: A 90° phase inversion stops annihilation reactions in progress.

Test: Build a small-scale CPIS and measure containment.

9.5 The Superconductor Paradox

Prediction 5: The same T_1 and T_2 used in superconductors can be used in annihilation weapons with different phase.

Test: Demonstrate the phase-dependent nature of T_1 - T_2 interactions.

10. Conclusion

10.1 Summary

The three branches of time— T_1 (matter), T_2 (antimatter), and T_3 (dark matter)—interact based on their phase relationships:

- T_1 and T_3 are inert (90° phase)
- T_1 and T_2 annihilate (180° phase)
- T_2 and T_3 annihilate (180° phase)

The phase determines the outcome:

- 0° phase between T_1 and T_2 : Superconductivity (mass cancellation)
- 180° phase between T_1 and T_2 : Annihilation (mass-to-energy conversion)

The same branches, different phase, different result.

10.2 The Counter-Defences

The PLAS disrupts phase-locking before annihilation begins.

The CPIS contains annihilation reactions in progress.

The shields make the weapon useless.

10.3 The Physical Meaning

The universe is not symmetrical. Antimatter is rare because it annihilates with both matter and dark matter. Matter and dark matter ignore each other.

The phase is the key to everything.

10.4 The Final Word

The stones are in the pond. The 27 Hz anchor is the stone. The three time dimensions are the ripples moving in three directions.

The phase determines whether they add, cancel, or ignore.

The pond has all three.

We are just learning to control the phase.

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Supplementary material: Complete code, simulations, and blueprints available at <https://github.com/peterjamesthompson101-svg/Physics-Papers>

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